

# NLP based text classification using TF-IDF enabled fine-tuned long short-term memory: An empirical analysis

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## ABSTRACT

The rapid proliferation of information through digital transformation and the widespread use of social networking platforms has significantly increased the speed of information dissemination across urban and rural areas alike. While these platforms have become vital channels for sharing news, advertisements, and crucial updates, they also pose challenges in verifying the authenticity of the information in real-time. Addressing this issue, this study proposes a novel Convolutional Neural Networks (CNN)-Long Short-Term Memory (LSTM) model designed for the classification of fake news articles. A comprehensive dataset covering diverse categories, including government news, Middle East news, US news, left-wing news, and political content, was utilized in this research. Following preprocessing, features were extracted using the Term Frequency-Inverse Document Frequency (TF-IDF) technique, and word embeddings were generated for enhanced semantic representation. The combined CNN-LSTM model leverages the strengths of both architectures, capturing local patterns and long-range dependencies within the data. The experimental results demonstrate that the Fine-Tuned CNN-LSTM model outperforms all precedent approaches across various categories. Notably, the Fine-Tuned CNN-LSTM model achieves the highest accuracy (AC), ranging from 0.57 to 0.68, highlighting its superior classification performance to other precedent approaches, indicating their inefficacy in handling multiple categories.

## 1. Introduction

On social media platforms like LinkedIn, Twitter, Facebook, and Instagram, information is widely shared, and these platforms have become the strong voice of the people. They have transformed and expanded the speed at which information is conveyed across society. Nonetheless, it has been accompanied by serious obstacles caused by false and misleading material.

The misleading content not only plays with people's emotions but also influences people to make wrong opinions and even puts human lives in danger [1]. It also raises the likelihood that the public would be misinformed about numerous subjects via conventional or social media. Thus, false news may be harmful and have enormous implications in the actual world [2]. Statistics show that the high proportion of misinformation and fake news is deteriorating society's peace and even the norms and rituals. Since internet information can be quickly generated, stored, structured, shared, and used, statistics

reveal that a large proportion of false information and fake news erodes society's peace, even norms and rituals. In a big data environment, fake news misleads people to make wrong judgments and negatively affects society. Text categorizations are based on TF-IDF and Word2vec for identifying fake news and can be validated using various conventional classification approaches like Support Vector Machines (SVM), Decision Trees (DT), Neural Networks (NN), etc. [3]. However, compared with past decades, social media has helped a lot to update information but has come with many challenges. In such a scenario, emerging areas like Machine Learning (ML) and Deep Learning (DL) are critical in facing the challenges of fake news, misinformation, and media manipulations. The latest innovation in ML gives more profound knowledge based on the previous history. It addresses the mechanisms and strategies to overcome the root cause of fake news through AI and DL. One potential challenge to misleading information is using DL-based text classification systems for fake news identification [4].

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Several researchers use deep Neural Networks to identify false news since they provide the highest performance accuracy in several areas, like text categorization, text summarization, and image and audio classification. As black-box models, neural network models rarely offer transparency and explainability when forecasting outcomes. In such a circumstance, Local Interpretable Model-agnostic Explanation-Bi-directional Long Short-Term Memory (LIME-BiLSTM) models assure the transparency and explainability of model prediction by retaining accuracy [5]. To account for each unique prediction, the LIME-BiLSTM technique, which refers to LIME, emulates a black box decision model with a local, Bi-LSTM was purposefully created to address the issue of long-term reliance. Their regular activity over long time frames is to recall information. LSTM learns significantly more quickly and produces numerous successful runs [6]. Moreover, it resolves challenging problems with artificially created lengthy time lags that earlier recurrent network methods cannot resolve. It converts documents to feature vectors using the Bidirectional Encoder Representations from Transformers (BERT) approach as a feature extractor. It is taken as inputs to train using a DL-based classification network. Similarly, XLNet, a generalized autoregressive pretraining method, enables the prediction of binary and multi-class problems in fake news recognition [7].

A Twitter survey found that false information spreads 100 times faster than verified content. Millions of individuals in certain countries may be impacted by anomalies caused by false news, although these impacts may be trivial [8]. In the proposed study, the LSTM model is trained using DL. The TF-IDF feature extraction mechanism is implemented after the dataset pre-processing phase, followed by training the model. The model's training progress is monitored on different epoch values and learning rates. The loss values are assessed for fine-tuning the model. The main contributions of the article are listed as follows.

- A novel fine-tuned CNN-LSTM model for classifying fake news articles is proposed. Specifically, the proposed Fine-Tuned CNN-LSTM Model first uses the TF-IDF weighting technique for feature extraction. Then, an embedding word layer is embedded to construct word vectors from vocabulary words. After that, the blended CNN and LSTM models capture all local and long-distance characteristics from the fake news dataset. Finally, the softmax classifier performs the multi-class classification task.
- The proposed model is evaluated using a publicly available fake news dataset and compared with state-of-the-art approaches that are being commonly used in fake news classification. To assess the efficiency of the proposed model.
- Feature extraction technique, i.e., TF-IDF implementation on the pre-processed dataset, improves performance and increases the LSTM model's learning rate.
- To train the model for the longer sequential text values and regularly monitor the training progress of the LSTM model behavior using optimization techniques and hyperparameters taken in the study.
- Implementing the Exploratory Data Analysis (EDA) analysis for classifying news articles using ML techniques and their validation accuracy compared with LSTM trained model.

**Impact Statement**— The current study presents a fine-tuned CNN that works with the LSTM component. In contrast to the conventional CNN models, the proposed approach performs the pre-processing of the data, which includes tokenization, text conversion, and stopword elimination. The pre-processing assists in the precise processing of the required data. The input is fed as a numerical sequence to the CNN, and the features are identified using the TF-IDF model. The model's efficiency at various epochs with varied learning rates for multiple classes is analyzed. Furthermore, the model's performance is deeply analyzed with other baseline models. The manuscript also outlines the dataset details, implementation environment, and initialization parameters to make it feasible to reproduce the results.

The current work explores multiple tiers of text classification models, beginning with TF-IDF-based feature extraction combined with traditional machine learning classifiers, progressing to a BiLSTM deep learning model, and culminating in a fine-tuned CNN-LSTM hybrid. The comprehensive comparative evaluation, including performance metrics like precision, recall, F1-score, and AUC across various real-world news categories, illustrates the incremental gains offered by the CNN-LSTM architecture. These results, particularly the superior accuracy and robustness of CNN-LSTM over BiLSTM and traditional models, demonstrate the significance of convolutional layers in extracting spatial text features before temporal modeling. While our approach may not leverage transformer-based embeddings, work offers valuable insights into the effectiveness of hybrid architectures on complex datasets, especially for scenarios with limited computational resources or when explainability of layered deep models is prioritized. The fine-tuned CNN-LSTM architecture demonstrates superior performance, particularly in terms of accuracy and AUC, achieving a validation accuracy of 81% over 50 epochs and consistently outperforming BiLSTM across various news categories such as Government, Middle-East, and Politics. Notably, the CNN-LSTM model achieved higher precision, recall, and F1-scores, reflecting its strength in capturing both local features and sequential dependencies. These findings highlight the effectiveness of integrating convolutional layers with LSTM networks and reaffirm the relevance of structured benchmarking, even in the context of complex real-world datasets.

The remaining paper is arranged as follows: Section 2 outlines the recent literature review on fake news articles. Section 3 outlines the most common precedent techniques used to perform classification. Section 4 discusses the proposed model and its functional component. In Section 5, experimental findings and results are analyzed. Finally, Section 6 concludes the paper by outlining future research directions.

## 2. Literature review

Fake news reviews come into the category of deceptive content, also known as opinion spam. The authors propose an n-gram model to detect fake content. Microblogging sites like Twitter and LinkedIn, part of social media, create vast amounts of data. It results in concentrating research on Opinion Analysis and text message exploration tactics [9]. The authors did a fake news text classification with deep learning approaches and made a prediction model in Python. Semantic approaches characterized the data using on-term frequency distribution and inverse document frequency weighting. The authors use eight machine-learning algorithms in the measuring and evaluation step, including Naive Bayesian (NB), AdaBoost, K Nearest Neighbor (KNN), Random Forest (RF), Logistic Regression (LR), DT, Neural Networks (NN), and SVM, as well as DL models like CNN, LSTM, Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU) [10].

Andre Rusli et al. [11] used a binary text classification strategy to identify false news items and distinguish them from legitimate ones using a supervised approach based on the Multi-Layer Perceptron (MLP). To combat misleading news and disinformation campaigns, ML has become one of the active lines of research that can classify text into fake or not fake [12]. The authors describe a framework named BerConvoNet, to categorize supplied news material as fake or authentic with little error. BerConvoNet has two major components: a News Embedding Block (NEB) and a Multi-Scale Feature Block (MSFB). NEB extracts word embedding from a news item using BERT [13].

Explainability is one of the challenges in the case of NN. Several factors must be considered while introducing explainability to a neural network. The text's length, the pre-processing technique, and the model's complexity all play a role in effective categorization [14]. Herley Shaori Al-Ash et al. [15] used an ensemble learning strategy to classify Indonesian false news. They compared the findings of the ensemble classifier, the RF classifier, to those of non-ensemble classifiers such as NB and SVM. Concentrating on content-based approaches for

identifying false news, such as binary text classification, is also helpful. The authors used a semi-supervised graph-based false news recognition algorithm based on Graphical Neural Networks (GNN) [16].

Apart from ML, DL techniques, Natural Language Processing (NLP), and feature extraction provide solutions to detect whether the small piece of text is fake news or not [17]. Fake news identification may also be accomplished via n-gram analysis and machine learning algorithms. Hadeer Ahmed [18] carried out the experimental evaluation using TF-IDF as a feature-based strategy, combined with the SVM classifier, which is the most effective method.

Okoro et al. [19] developed a Machine-Human model to trick people on social media into believing false information. The authors looked at three datasets to compare and contrast the features of fake and real news using Glove Integrating and Character Embedding [20]. Ozbay and Alatas [21] focus on supervised classifiers in their study of text classification methods. Using a refined set of linguistic factors, Gravanis et al. [22] examined successful classification approaches on five separate datasets of false and factual news to determine which were more effective at identifying the former.

For text augmentation, fusing evolutionary algorithms and graph-based neural networks can produce better text data. Genetic algorithms improve data conversion, whereas graph-based models record associations amid words, improving context safety. This interplay can improve model act and training in natural language processing tasks [23]. BERT-based dynamic fusion is the combination of dynamic fusion methods and BERT (Bidirectional Encoder Representations from Transformers) embeddings. These methods entail the flexible fusion of data from several sources or procedures. How information is merged throughout this fusion process may vary depending on the task or input [24]. A method employed in situations of unbalanced learning is consensus clustering-based undersampling. It entails applying clustering on samples from the majority class to create several subsets of the majority class. To create balanced or less unbalanced training sets, these clusters are then joined, and the classification model is trained using these sets [25].

Word embeddings and clustering are used in the 2-stage topic extraction approach for bibliometric data analysis to find subjects within a set of bibliographic data. The words in the documents are first encoded using word embeddings, which show words as dense vectors capturing semantic associations. These embeddings make it possible to better serve topics by capturing the context and meaning of words. The word embeddings are subjected to the second step, where clustering techniques are used to group related words together [26]. Sentiment categorization is one of the techniques that can be improved via a hybrid ensemble pruning method based on multi-objective evolutionary algorithms and consensus clustering. Consensus Clustering is a technique that creates numerous clusters from the training data, allowing for various interpretations of how the information should be organized. These groupings show possible subgroups of the ensemble's members. Multi-Objective Evolutionary Algorithm optimizes the ensemble by taking into account many goals at once, such as classification accuracy and member dissimilarity [27].

While recent advances have focused heavily on transformer-based architectures and contextual embeddings, this study emphasizes the continued value of integrating convolutional and recurrent layers for structured text classification. By combining CNNs for local feature extraction with LSTMs for sequential learning, our fine-tuned CNN-LSTM model offers a balanced approach that outperformed both traditional machine learning models and BiLSTM baselines. Although the work does not incorporate transformer models or contextual embeddings like BERT, it demonstrates that a well-optimized hybrid model can still achieve strong results such as an 81% validation accuracy on real-world, multi-class datasets. This establishes a practical, interpretable, and resource-efficient alternative for domain-specific classification tasks.

### 3. Background

The current section discusses and summarizes the most common precedent techniques found in the literature for classifying fake news articles, detailed as follows.

#### 3.1. Precedent approach 1: Decision tree

DT-based classifiers have been intensively studied for fake news detection. For instance, [28] used a DT algorithm to detect fake news from a dataset of 20,761 samples, with a testing sample size of 4,345. In the Pre-processing steps, the authors removed unnecessary characters and stop words. The DT model obtained an accuracy of 89.11%. By using the DT C4.5 classification strategy and a dataset of 50,610 tweets, another study [29] attempts to detect fake news on Twitter. The n-grams produce the features, and Information Gain is used for feature selection. Classification testing using weighting significant features and extracting features with a ratio of 90:10 split of the dataset for training and 5000 unigram features yielded the greatest accuracy rate of 72.91%. DT is a tree structure that shows predictions using factor-based division. It starts with a root node and ends with a decision at the leaves. The node from which it is subdivided into branches is called a root node. The Gini index (GI) is to be calculated as shown in Eq. (1) to calculate the feature for the root node. The feature with the lowest impurity will be selected as the root node. Gini Index and Entropy are the criteria for determining the information gain and splitting a node. Entropy may be calculated for a dataset containing multiple classes, as shown in Eq. (2).

$$GI = 1 - \sum_{i=1}^n (p_i)^2 \quad (1)$$

$$E = - \sum_{i=1}^n p_i \log_2 p_i \quad (2)$$

In the above equation, the notation  $E$  represents the entropy and  $p_i$  designates the probability of randomly selecting an example in class  $i$ .

#### 3.2. Precedent approach 2: Naïve Bayes

The use of NB-based classifiers for fake news recognition is among the widely researched areas. Particularly, the article uses [30] NB algorithm for the fake news recognition problem. The research results indicate that using NB with n-gram slightly increases the performance of both TF-IDF and Count Vectorizer in detecting fake news. The research found that the TF-IDF Vectorizer has a better classification precision in recognizing fake news. In contrast, the Count Vectorizer is efficient in differentiating fake and real news in a relatively balanced manner. Another article [31] uses NB and LSTM classifiers to identify fake news effectively. The experiment's outcome shows that the LSTM classifier outperforms the NB classifier. NB is largely inspired by the Bayes theorem and probability estimates. NB works well for predicting multiple classes. For a given dataset, there are input variables and a target variable. NB converts the training dataset into a frequency table, assuming all the features that predict the target values are separable. It approximates the probability at each class. It assesses the probability of an event concerning prior insight into the conditions that might be related to it. For a feature set  $y = y_1, y_2, \dots, y_n$  and a target variable  $n$ , states in Eq. (3).

$$p(n|y) = \frac{p(y|n) \times p(n)}{p(y)} \quad (3)$$

NB picks the hypothetical value with the highest probability, known as posterior probability, shown in Eq. (4).

$$p(n|y_1, \dots, y_n) \propto p(n) \pi_{i=1}^n p(y_i|n) \quad (4)$$

### 3.3. Precedent approach 3: Support vector machine

SVM has indeed been extensively researched in detecting fake news [32,33]. It is a supervised learning approach that classifies data by locating the best hyperplane and optimizing the margin across classes. SVM is effective in detecting fake news by identifying complex relationships between the features and labels. In SVM, there is a separating hyperplane for separating the classes into positive and negative examples and the objective function for separating the hyperplane in the form of an Eq. (5). Here,  $w$  is normal to the hyperplane. Class labels, i.e.,  $y \in -1, 1$  are implied as  $-1$  for weak class and  $+1$  for decisive class in SVM.

$$\pi : w^t x + b = 0; \quad (5)$$

The functional limit of a hyperplane for the  $i$ th training example is defined in Eq. (6), and the functional margin of a hyperplane concerning the complete dataset is defined in Eq. (7).

$$\gamma^i = y^{(i)}(w^t x^{(i)} + b) \quad (6)$$

$$\gamma = \min_i \gamma^i \text{ for } i = 1, 2, \dots, m \quad (7)$$

### 3.4. Precedent approach 4: K nearest neighbor

KNN is another classification algorithm that has been explored for fake news detection. For instance, [34–36] research studies use the KNN for fake news detection and its performance is promising. A supervised learning approach classifies data based on its neighboring samples. It includes a non-parametric technique that identifies data points over the training set depending on the proximity to surrounding points. KNN classifies all new cases based on the similarity measure. The distance among two points  $(x_1, y_1)$  and  $(x_2, y_2)$  within the plane with coordinates in KNN is assessed using the Euclidean distance given in Eq. (8).

$$d(x_1, y_1), (x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (8)$$

After aggregating the distance, the input  $x_2$  gets defined to the class with the greater probability, as shown in Eq. (9).

$$p(Y = i | X = x_2) = \frac{1}{N} \sum_{j \in \alpha_0} I(y_j = i) \quad (9)$$

In the equation,  $X$  is a collection of features from a sample, and  $Y$  is a class label for the provided dataset. Now, KNN is an approach to classification that assigns a data point  $x_2$  to the class with the highest certainty after estimating the conditional distribution of  $Y$  given  $X$ . Using the above equation, where  $\alpha_0$  is the set of  $N$ -nearest observations and  $I(y_j = i)$  is a measurement variable that evaluates to 1 if a given observation  $(x_j, y_j)$  in  $\alpha_0$  is an instance of class  $i$ , and 0 contrary, KNN now examines the  $N$  observations closest to a test observation  $x_2$ . And figures the conditional probability that it relates to class  $i$  provided an integer with a positive value  $N$ . Here for the distance measure  $i$  value is to be selected. If  $i = 1$ , then it is Manhattan Distance; if  $i$  value is equal to 2, it is Euclidean Distance. The corresponding formulas for Euclidean Distance is shown in Eq. (10) and Manhattan Distance is shown in Eq. (11).

$$I_1(x, y) = \sqrt{\sum_{j=1}^n (x_j - y_j)^2} \quad (10)$$

$$I_2(x, y) = \sum_{j=1}^n |x_j - y_j| \quad (11)$$

### 3.5. Precedent approach 5: Random forest

RF is another popular classification algorithm that has gained attention in fake news detection research, including [37], which uses a counting vector and a Repetition Inverse Document (RID) matrix to identify fake news. However, these methods do not consider the arrangement and context of words, which can lead to inaccuracies in classification. It creates multiple DTs and combines them to improve the overall accuracy and reduce overfitting. This technique uses ensemble learning, blends many weak classifiers, and boosts performance. It consists of many DTs and takes the prognosis from each tree rather than depending on one tree and, based on the maximum votes of predictions, predicts the final output. Here, feature extraction is important and is considered the decline in entropy intact by the possibility of ending at that node. The node interest is calculated using the Gini Index, as shown in Eq. (1). However, the emphasis of each attribute on a DT is forecasted as shown in Eq. (12). Here,  $attrib_{ji}$  shows the concern of attribute  $j$  and  $j^{n_{ji}}$  represents the importance of node  $i$ .

$$attrib_{b_j} = \frac{\sum_i : \text{node } i \text{ splits on attribute } attrib_{j_i}^n}{\sum_{x \in \forall \text{ nodes } n_{j_x}} n_{j_x}} \quad (12)$$

The above equation is distributed to a value between 0 and 1 by dividing by the sum of all importance feature values, as shown in Eq. (13).

$$\text{normalized}(attrib_{j_i}) = \frac{attrib_{j_i}}{\sum_{i \in \forall} attrib_{j_i}} \quad (13)$$

As indicated in the equation, the important value of the characteristic is derived by adding the values for each tree and dividing by the sum of trees as shown in Eq. (14). Here,  $rf \text{ } attrib_{j_i}$  presents the influence of attribute  $j$  approximated from all trees in the RF, and  $\text{normalized } attrib_{j_i}$  is the normalized feature significance for  $j$  in tree  $i$ , where the notation  $T$  designates the number of all trees.

$$rf \text{ } attrib_{j_i} = \frac{\sum_{i \in \forall} \text{normalized } attrib_{j_i}}{T} \quad (14)$$

### 3.6. Precedent Approach 6: BiLSTM

As a baseline deep learning approach, BiLSTM network was implemented for comparative analysis. BiLSTM is well-suited for sequential data modeling, as it processes input in both directions, forward and backward, capturing past and future contextual dependencies. In the context of this study, the BiLSTM model was trained using the same hyperparameters and dataset partitions as the proposed CNN-LSTM framework to ensure a fair comparison. While BiLSTM demonstrated a reasonable ability to differentiate between various news categories, it consistently yielded lower AUC values across most classes. Specifically, its classification performance was found to be suboptimal for categories like "Govt" and "US\_News," indicating limited capability in capturing intricate spatiotemporal features. These findings underscore the enhanced learning capacity of the proposed CNN-LSTM model, which integrates convolutional layers for spatial feature extraction with LSTM layers for temporal dynamics, resulting in superior classification performance. Formally, for a given input sequence  $X = [x_1, x_2, \dots, x_T]$ , a BiLSTM computes two hidden states: a forward hidden state  $\vec{h}_t = LSTM \{x_t, \vec{h}_{t-1}\}$  and a backward hidden state  $\overleftarrow{h}_t = LSTM \{x_t, \overleftarrow{h}_{t-1}\}$  which are concatenated as  $h_t = [\vec{h}_t; \overleftarrow{h}_t]$  to form the final output representation at time  $t$ . The output is formulated as  $H = h_1, h_2, \dots, h_T$  which is denoted by the hidden state, and the classification procedure is involves a softmax layer as shown in Eq. (15).

$$\hat{y} = \text{softmax}(\omega_0 h_T + b_0) \quad (15)$$

From the above equation, the  $\omega$  is associated with the weight and the notation  $b$  denoted the bias value. which are being updated over the epochs to minimize the loss. This dual-context representation



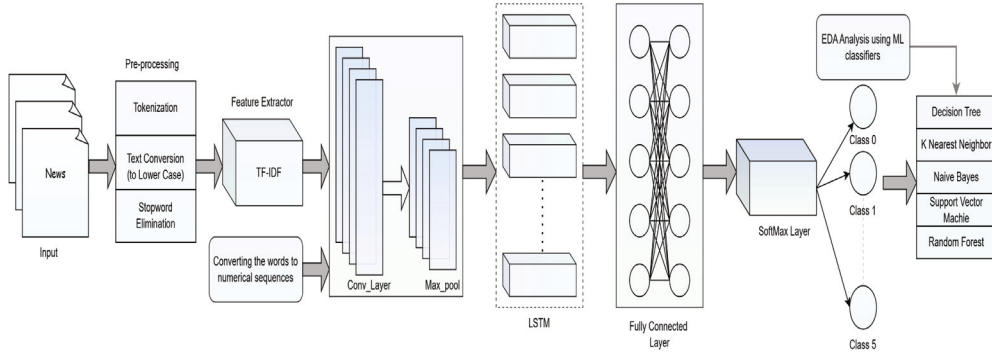


Fig. 1. Proposed Fine-Tuned CNN-LSTM model for classifying fake news articles.

allows the model to capture dependencies that may be missed in unidirectional setups, which is particularly useful for complex textual semantics found in news datasets. In our study, the BiLSTM model was trained using categorical cross-entropy loss and optimized via the Adam optimizer. While it demonstrated balanced learning and handled temporal dependencies well, the absence of localized feature extraction typically offered by convolutional layers limited its classification strength. Compared to our proposed CNN-LSTM architecture, which first applies convolutional layers to highlight spatial hierarchies before sequential modeling, the BiLSTM lagged in key performance metrics such as AUC and F1 score. These results highlight the advantage of combining convolutional operations with sequential learning for more discriminative feature representations in multiclass text classification tasks [38].

#### 4. Proposed framework

This section discusses the functional components used to design the proposed framework. The procedure for using the Fine-Tuned CNN-LSTM model for classifying fake news articles consists Pre-Processing Phase and the Fine-Tuned CNN-LSTM Model, as shown in Fig. 1. The following subsections give a detailed explanation of each component.

The figure outlines a complete text classification framework, combining both deep learning strategies and conventional ML methods. The initial raw text inputs are first processed through a series of steps, which include tokenization, converting all text to lowercase, and the elimination of common stopwords. The processed text is converted into numerical forms utilizing methods like TF-IDF and word embeddings. These features are fed into a CNN layer, then max pooling is applied to capture local patterns in the text. Following this, an LSTM network processes the sequential data in order to retain contextual dependencies. The output from the LSTM network is passed into both a fully connected layer and a Softmax layer to generate probability scores for each class. The EDA module uses several ML classifiers to conduct further analysis and compare performance across multiple target classes.

##### 4.1. Pre-processing phase

There are a lot of outliers and noise in the raw news articles dataset or the data fetched online. The data is pre-processed to remove specific noisy terms and insignificant data to ensure precision. This includes text tokenization, text conversion to lowercase, and stopword elimination. In text tokenization, the statements are divided into words called tokens. Although these tokens are typically referred to as terms or phrases, it is important to distinguish between them. A sequence of characters known as a functional semantic unit in a text is an example of a token. Each token has a type, even those with the same string of characters. Word categories are included in an information retrieval system's vocabulary. As a result, the text sequences are tokenized into

tokens. The text is then changed to lowercase, which creates a harmonized string using the same word in both uppercase and lowercase. Finally, stop words such as “an”, “some”, and “we” are eliminated from the datasets to enhance the classifier's efficiency. The statistical information of the dataset is an essential part of pre-processing, and for a better understanding of the trends present in the dataset. In the Pre-processing phase, the text is tokenized, then converted to lowercase, and punctuation is removed from the text.

##### 4.2. Fine-tuned CNN-LSTM model

In this subsection, we have discussed the various components used to design the proposed Fine-Tuned CNN-LSTM Model.

###### 4.2.1. Feature extraction using TF-IDF

After the pre-processing phase, the TF-IDF weighting technique extracts features. It is one of the statistical methods in NLP for information retrieval. It is a method of text vectorization in which the significance of individual words in a document is accentuated using numerical values. The first step in NLP (text data) is to tokenize the data. It is essential because the context of the document could be swiftly translated based on an interpretation of the terms used in the language [39]. When comparing one source to a set of documents, the TF-IDF metric assesses the relevance or significance of string lexicons such as words, phrases, lemmas, and so forth. When extracting and grading the relevance of a text given a keyword, search engines frequently use variations of the TF-IDF weighting method [40,41]. With TF-IDF, a term is determined as the product of TF and IDF values, as shown in Eq. (16).

$$tf\_idf = tf_{x,y} \times idf(\omega) \quad (16)$$

In the above equation,  $tf$  is the term frequency, and the proportion of word frequency in a document is measured as the sum of words in that document, as shown in Eq. (17).  $n_{x,y}$  is the sum of times a word occurred in a document, i.e., occurring  $x$  in  $y$ .

$$tf_{x,y} = \frac{n_{x,y}}{\sum_{i \in y} n_x(i)} \quad (17)$$

$$idf(\omega) = \log \frac{D}{df_x} \quad (18)$$

In the above Equation,  $idf$  represents the inverse data frequency and is used to assess the weight, i.e.,  $\omega$ , of the words that rarely occur across all the documents in the dataset and is mathematically shown in Eq. (18). As shown in Eq. (19), the TF-IDF equation can be calibrated, where the variable ‘D’ represents the total number of documents,  $tf_{x,y}$  is the number of occurrences of  $x$  in  $y$  and  $df_x$  designates the sum of documents containing  $x$ .

$$\omega_{x,y} = tf_{x,y} \times \log \frac{D}{df_x} \quad (19)$$

#### 4.2.2. Word embedding layer

Word embedding approaches encode words as real-valued vectors in a common vector space. Since mapping words to vectors and learning their values is conceptually similar to training a neural network, this approach is often used in deep learning. For every word, the technique is primarily built around a distributed approach. A real-valued vector with hundreds of values is used to represent each syllable. Sparse word samples, such as those used in one-hot encoding, need hundreds of thousands of dimensions. The common meaning is conveyed via word choice. That enables semantically identical representations of words used in similar circumstances.

#### 4.2.3. Convolutional layer

The convolution layer computes the properties of the input data after filtering it to obtain features. The convolution layers in NLP applications are set up as parallel layers that work on document matrices simultaneously. To extract features at each place, convolution filters are applied to the document matrix by swiping linear filters over the text. Several filters with different window sizes are used to extract numerous features. The convolution operation has the following mathematical form as shown in Eqs. (20) and (21).

$$x_{ij}^1 = \sigma \left( b_j^1 + \sum_{m=1}^M w_m^1 \times j^{x_i^0} + m - 1, j \right) \quad (20)$$

$$y_{ij}^1 = \sigma \left( b_j^1 + \sum_{m=1}^M w_m^1 \times j^{x_i^0} + m - 1, j \right) \quad (21)$$

where  $x_{ij}^1$  means the output of layer 1;  $j^{x_i^0}$  is the input of layer 1;  $w_m^1$  is weight parameter of the convolution kernel;  $b_j^1$  represents the bias for the  $j$ th feature;  $m$  denotes the index value of the filter;  $\sigma$  is activation function.

#### 4.2.4. Max-pooling layer

The convolutional layer's output is subsampled using the max-pooling layer. The max-pooling layer may save the most critical information by removing the local dependencies within distinct areas. This also helps in reducing the overfitting issue and is performed using Eq. (22).

$$p_{ij}^1 = \max_{r \in R} y_{i \times T + R, j}^{l-1} + R_j \quad (22)$$

Where  $R$  is the pooling size of less than the size of the input  $y$ , and  $T$  is the step determining how far to shift the input data area. The acquired region matrix is vectorized and sent to the LSTM-based sequential layer.

#### 4.2.5. LSTM-based sequential layer

The LSTM model is taken for news category classification in the proposed work. The discipline of DL relies on these kinds of networks. It is a kind of RNN that may be trained to solve issues requiring knowledge of the future, such as sequential prediction. LSTM makes little adjustments to the data by adding or multiplying across cell states. Unlike RNN, which modifies preexisting data by applying certain functions, it forgets and remembers values selectively. Memory cells, sometimes called cell states, are what LSTM comprises. In this case, the LSTM cell state may have data inserted or withdrawn using gates. It contains a layer of sigmoid neural nets and a multiplication operation. The output of the sigmoid layer is in the range of 0 and 1, as the name suggests. The value 0 means not to allow passing through, and the value 1 allows every piece of information to pass through.

The purposeful creation of LSTM was to address the issue of long-term reliance. Their primary activity over extended periods is essentially memory. Both recurrent neural networks are constructed from a series of redundant modules. This repeating module would have a relatively simple structure in regular RNNs, such as a single  $\tanh$  activation function [42]. A single LSTM cell comprises four distinct components: the Forget gate, the input gate, the output gate, and the

#### Algorithm 1 FINE-TUNED CNN-LSTM Model

---

```

1: procedure INPUT: (Read the dataset)
2:   OUTPUT: Classification outcome of the news articles
3:   Partition of the data for Training, Validation, and Testing.
4:   Visualize the training text data using a word cloud.
5:   Preprocess Text Data
      Tokenize text  $\rightarrow$  tokenizedDocument ()
      Convert text to lowercase  $\rightarrow$  lower()
      ErasePunctuation  $\rightarrow$  erasePunctuation()
6:   Convert Document to Sequences of numeric indices.
7:   Feature Extractor using TF-IDF technique
       $tf_{x,y} = tf_{x,y} \times idf(\omega)$ 
       $tf_{x,y} = \frac{n_{x,y}}{\sum_i n_{x,y}}$ 
       $idf(\omega) = \log \frac{D}{df_x}$ 
8:   Train and validate fine-tuned CNN-LSTM Model
      Add a Convolutional Layer and Max-Pooling Layer
       $x_{ij}^1 = \sigma \left( b_j^1 + \sum_{m=1}^M w_m^1 \times j^{x_i^0} + m - 1, j \right)$ 
       $y_{ij}^1 = \sigma \left( b_j^1 + \sum_{m=1}^M w_m^1 \times j^{x_i^0} + m - 1, j \right)$ 
       $p_{ij}^1 = \max_{r \in R} y_{i \times T + R, j}^{l-1}$ 
      Add LSTM Layers
       $f\_gate = \sigma_g(\omega_{Input}^1 \times x_2 + \omega_{prev}^1 \times h_{state} + b_f)$ 
       $i\_gate = \sigma_g(\omega_{Input}^2 \times x_2 + \omega_{prev}^2 \times h_{state} + b_i)$ 
      Add softmax layer
       $p(\hat{y}_c = y_c | x) = \phi(x)_{y_c} = \frac{e^{x_c}}{\sum_j e^{x_j}}$ 
      Calculate cross-entropy loss across the categorical data
       $L(\hat{y}_c, y_c) = - \sum_{i=1}^n \sum_{c=1}^C y_c^i \cdot \log(p(\hat{y}_{ic} = y_{ic} | x_i))$ 
9:   Perform Testing using Testset.
10:  Evaluate performance using various metrics
11: end procedure

```

---

cell state. Forget gate, as the name implies, this section determines whether or not information from the previous step will be discarded. The first sigmoid layer does this. In each time step  $t$ , it considers the current input and the hidden state of the cell and picks a range of 0 to 1 for each value in the cell's state. Everything is saved if the value is 1. Information is completely discarded when all bits are set to zero, but the amount of carryover from one state to the next is controlled by the values used in the bit pattern. The connection between these conditions is a dot product in Eq. (23).

$$y = x_1 \cdot \omega_1 + x_2 \cdot \omega_2 + x_3 \cdot \omega_3 \quad (23)$$

Here,  $x$  and  $w$  are the vectors. The dot product notation of the activation function is shown in Eq. (24).

$$f_x = \omega^T \cdot x \quad (24)$$

$\omega_1$ ,  $\omega_2$ , and  $\omega_3$  are called weights or learning training parameters. In the dot product, the hidden layer is the result of the dot product. An important property of the sigmoid activation function is that. Its range is between [0-1], which helps in binary classification problems, i.e., Yes or No type outcome.  $\tanh$  is mostly used for hidden layers and is almost similar to sigmoid, whose range is between [-1, 1], and makes the decision. Each cell serves as a memory, and gate states are used to manipulate each cell's contents [43].  $\tanh$  is taken as the data normalization function, and the present status is shown in Eq. (25). In Eq. (26),  $pres_{state-x_1}$  represents the present state. The  $pres_{state-x_1}$  becomes the previous state, as shown in Eq. (27).

$$pres_{state-x_1} = \tanh(\omega_{Input}^T \times x_1) \quad (25)$$

$$prev_{state} = pres_{state-x_1} \quad (26)$$

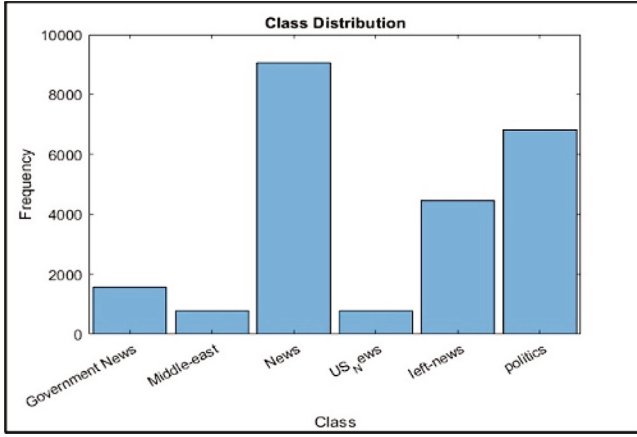


Fig. 2. Class distribution of the dataset.

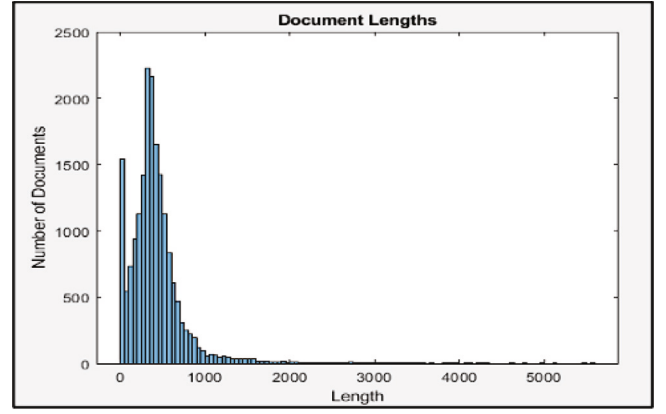


Fig. 3. Histogram of the training document lengths.

$$x_2 = \tanh(\omega_{Input}^T \times x_2 + \omega_{prev}^T \times prev_{state}) \quad (27)$$

For any given point  $x_t$ , the current state is assessed as shown in Eq. (28). LSTM model resolves the problem of not remembering the long-term dependencies, unlike RNN. LSTM has two hidden layers: the cell and the hidden states. The sigmoid activation function is represented in the form of forget gate, i.e.,  $f\_gate$ , and the input gate, i.e.,  $i\_gate$ , in Eqs. (29) and (30). In the equations,  $h\_state$  is the hidden state,  $\sigma_g$  is the gate activation function, and  $b_f$  and  $b_i$  denote the bias for forget and input gate.

$$pres_{state-x_t} = \tanh(\omega_{Input}^T \times x_t + \omega_{prev}^T \times prev_{state}) \quad (28)$$

$$f\_gate = \sigma_g(\omega_{Input}^1 \times x_2 + \omega_{prev}^1 \times h_{state} + b_f) \quad (29)$$

$$i\_gate = \sigma_g(\omega_{Input}^2 \times x_2 + \omega_{prev}^2 \times h_{state} + b_i) \quad (30)$$

#### 4.2.6. Softmax layer

The output layer, which consists of a softmax classifier, will receive the  $1 \times E$  vector. The Softmax classifier can convert input vectors from  $C$ -dimensional space into categories and then output the findings as probabilities. The corresponding mathematical formula is shown in Eq. (31).

$$p(\hat{y}_c = y_c | x) = \frac{e^{x_c}}{\sum_j^C e^{x_j}} \quad (C = 1, 2, \dots, c) \quad (31)$$

To reduce the average categorical cross-entropy loss, which measures the disparity between model predictions and assigned labels, the softmax classifier is trained using back-propagation. Hence, the softmax classifier's cost function (categorical cross-entropy loss) is given in Eq. (32).

$$L(\hat{y}_c, y_c) = - \sum_{i=1}^n \sum_{c=1}^C y_c^{x_i} \cdot \log(p(\hat{y}_{ic} = y_{ic} | x_i)) \quad (32)$$

In Algorithm 1, we have discussed the procedure to perform classification using the proposed fine-tuned CNN-LSTM model.

## 5. Results and discussion

The effectiveness of the proposed methodology is discussed in this section. For experimental analysis, a publicly available dataset on Kaggle named "Fake and real news dataset" is used [44]. This dataset consists of longer news titles and description features. The class distribution of the sample dataset, along with its frequency, is shown in Fig. 2. There are two main classes, where all the data samples

Table 1

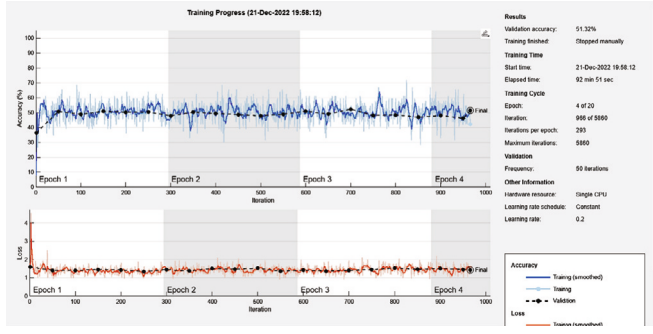
Simulation parameters taken for Fine-tuned CNN-LSTM model.

| Parameter                     | Value                     |
|-------------------------------|---------------------------|
| Training data                 | 18760                     |
| Tokenized Document            | 18760                     |
| Tokenized Validation Document | 4690 × 1                  |
| Solver Name                   | Adam Optimizer            |
| Encoding Scheme               | Word Encoding             |
| MiniBatchSize                 | 16                        |
| Layers                        | 6 × 1 convolutional layer |
| GradientThreshold             | 0.2                       |
| Embedding Dimension           | 50                        |
| No. of Hidden Inputs          | 80                        |
| Shuffle Value                 | Epoch                     |
| Number of Classes             | 6                         |
| Number of words               | 101 637                   |
| Sequence Length               | 500                       |

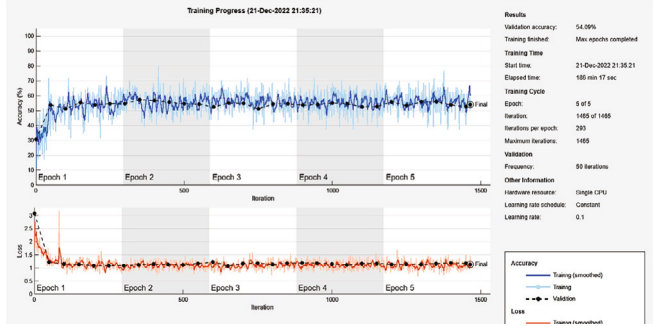
are labeled as True or Fake. The total number of instances in the "Fake" news dataset is 23481, and 21417 instances of true news. There are 4 attributes: Title, Description, Category, and Date. The dataset contains multiple classes, and there are 6 categories of news. The target feature 'Category' is converted into the numerical form by using the Label Encoder approach and fit into the data frame to achieve better performance of ML algorithms and for fine-tuned CNN-LSTM model training. Apart from it, 'NA' values are removed from the dataset by a text value to save more data for training. In Table 1, several hyperparameters, including the tokenized document, encoding scheme, layers, and many more, are listed along with their values. This table can be utilized to map with the proposed fine-tuned CNN-LSTM model for a comprehensive understanding. Table 2 discusses the implementation environment to execute the proposed framework. Once the dataset is pre-processed, it is bifurcated into training and testing in a 70:30 ratio, which the proposed fine-tuned CNN-LSTM model uses for classifying the fake news articles. The work is extended and compared with well-known precedent classifiers like NB, SVM, KNN, RF, and DT algorithms under a multi-class classification strategy.

### 5.1. Evaluation metrics

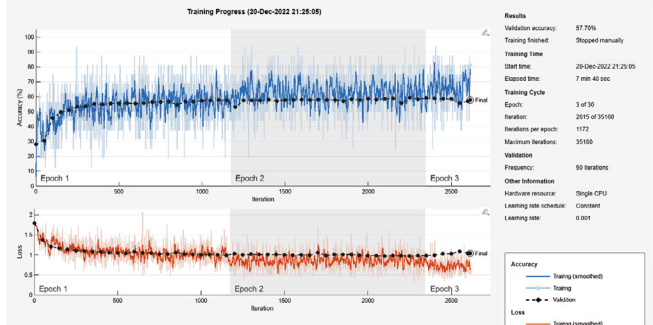
The formulas for the metrics that are used in the current study are formulated using measures like true positive (Trpo), which determines the number of correctly identified records as fake news, and true negative (Trne), which determines the sum of correctly recognized instances of genuine news as non-fake. The other two measures, like False Positive (Fapo), which determines the sum of instances that the model misinterprets as fake news as non-fake news, and False Negative (Fane), which determines the non-fake cases that are recognized as fake news. Accuracy (AC) is the proportion of correctly classified data



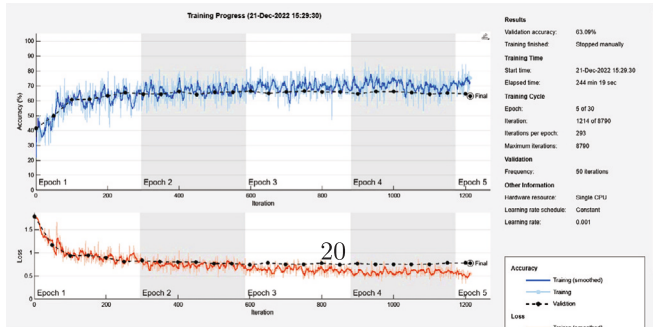
(a) Validation Accuracy of 51.32% on iteration 966, epoch value 4 of 20, and learning rate 0.2



(b) Validation Accuracy of 54.09% on iteration 1465, epoch value 5 of 5, and learning rate 0.1



(c) Validation Accuracy of 57.70% on iteration 2015, epoch value 3 of 30, and learning rate 0.001



(d) Validation Accuracy of 63.09% for iteration 1214, epoch value 5 of 30, and learning rate 0.001

**Fig. 4.** Result interpretation (Part 1) in terms of validation accuracy for Fine-Tuned CNN-LSTM model.

instances over the entire data instances. The corresponding formula is shown in Eq. (33). The metric sensitivity in foreseeing successful

**Table 2**

Details of implementation environment.

| System particulars     | Specification                                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating System       | Microsoft Windows 10                                                                                                                                                                                  |
| Architecture Type      | x64-based Instruction set                                                                                                                                                                             |
| Processor              | 12th Gen Intel (R) Core <sup>TM</sup> , i7-1255U@4.70 GHz, 3200 MHz, 4 Core (s), 10 Logical Processor (s)                                                                                             |
| Installed Physical RAM | 8 GB                                                                                                                                                                                                  |
| Computing Environment  | MatLab R2022b with Statistical and Machine Learning Toolbox, Deep Learning Toolbox, and Text Analytics Toolbox, Python 3.11 with Jupyter Notebook 6.4.12 (Anaconda3-2022.10) Windows-x86_64 Installer |

cases is measured using a precision measure (PR). It is determined by dividing the total number of true positives (Trpo) by the total number of true positives (Trpo) and false positives (Fapo) as shown in Eq. (34). Recall (RC), also known as sensitivity, measures how many genuine samples have been optimistic that an ML model correctly categorized. The number of genuinely positive (Trpo) instances divided by the sum of all positive examples (Trpo) and the number of false-negative (Trne) instances is the recall rate. It evaluates how well the machine learning model can interpret the data and pinpoint the real outcome. The corresponding formula is shown in Eq. (35). The F-1 Score quantifies the harmonic mean of memory and precision. It is beneficial for assessing algorithms for binary categorization issues. The corresponding formula is depicted from Eq. (36).

$$AC = \frac{Trpo + Trne}{Trpo + Fapo + Trne + Fane} \quad (33)$$

$$PR = \frac{Trpo}{Trpo + Fapo} \quad (34)$$

$$RC = \frac{Trpo}{Trpo + Fane} \quad (35)$$

$$F1 = 2 \times \frac{RC \times PR}{PR + RC} \quad (36)$$

## 5.2. Simulation analysis

The simulation results showed that most training materials include less than 500 tokens, as shown in Fig. 3. The training materials are translated to a series of numeric indices, with the token length set to 500. The document is converted into sequences before training the fine-tuned CNN-LSTM model, followed by the step to pad and truncate documents to similar lengths. Using padding and reducing word vector sequences may optimize training. The sequence input layer is built in the fine-tuned CNN-LSTM model. The input size is adjusted to 1, accompanied by a word embedding layer, i.e., words with comparable meanings are given the same numerical value. The model's training progress is monitored, and the validation accuracy is obtained for the different epochs, as shown in Figs. 4 and 5. The accuracy achieved is 51.32% with an epoch value of 20. The initial learning rate value is set to 0.2, shown in Fig. 4(a). In Fig. 4(b), the model's accuracy is 54.09% using an initial learning rate value of 0.1. The fine-tuned CNN-LSTM model achieved an accuracy of 57.70% for epoch value 3. The learning rate value set to 0.001 is shown in Fig. 4(c). The model achieved an accuracy of 63.09% for iteration value 1214 and maximum epoch value 5 out of a maximum 30, as shown in Fig. 4(d) on manual stopping, as the model took more elapsed time for training.

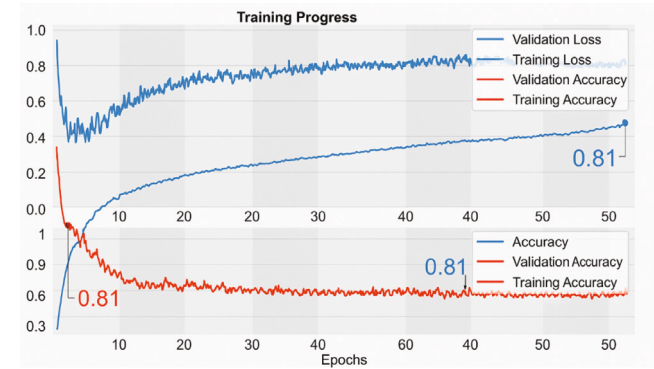
The training progress of the CNN-LSTM model is visualized through the plotted curves of training loss, validation loss, and accuracy across 50 epochs. As seen in the figure, the model demonstrates a consistent decline in both training and validation loss, indicating effective



**Table 3**

Analysis on various performance metrics for Fine-Tuned CNN-LSTM model.

| Validation accuracy | Elapsed time | Epoch   | Iteration | Iterations per epoch | Max iterations | Frequency     | Hardware resource | Learning rate |
|---------------------|--------------|---------|-----------|----------------------|----------------|---------------|-------------------|---------------|
| 51.32%              | 92 min 51 s  | 4 of 20 | 966       | 293                  | 5860           | 50 iterations | Single CPU        | 0.1           |
| 54.09%              | 186 min 17 s | 5 of 5  | 1465      | 293                  | 1465           | 50 iterations | Single CPU        | 0.1           |
| 57.70%              | 7 min 40 s   | 3 of 30 | 2615      | 1172                 | 35 160         | 50 iterations | Single CPU        | 0.001         |
| 63.09%              | 244 min 19 s | 5 of 30 | 1214      | 293                  | 8790           | 50 iterations | Single CPU        | 0.001         |
| 81.23%              | 52 min       | 50      | 2450      | 49                   | 2450           | 50 iterations | Single CPU        | 0.001         |



(a) Validation Accuracy of 81.00% after 50 epochs and learning rate 0.001

**Fig. 5.** Result interpretation (Part 2) in terms of validation accuracy for Fine-Tuned CNN-LSTM model.

learning and minimal overfitting. The validation accuracy shows a steady improvement, ultimately reaching 81% by the 50th epoch. This performance trend highlights the model's capability to generalize well on unseen data while maintaining stability during training. The results affirm that the CNN-LSTM architecture efficiently learns complex patterns from the input dataset over time. The overall result interpretation for the proposed fine-tuned CNN-LSTM model regarding various evaluation metrics is shown in [Table 3](#).

### 5.3. Comparison with precedent techniques

In this current subsection, the performance of the proposed fine-tuned CNN-LSTM model is compared with dominantly used predecessor methodologies. The details of these precedents with their computational equations are mentioned in [Section 3](#). The model's performance is evaluated and compared in terms of the Area Under the Curve (AUC)-Receiver Operator Characteristic (ROC) graphs. It is among the decisive benchmarks for checking the model act for analysis. It also visualizes the performance of classifiers while dealing with the category classification of the news articles, which is a multi-classification problem in the proposed work. It also determines the models' capability to differentiate between the classes. The concern of the proposed model is the dependency on better hardware configuration and the longer time it takes to learn deeply and get trained. The ROC-AUC curve for categorizing the news articles by the proposed fine-tuned CNN-LSTM model and precedent techniques is shown in [Fig. 6](#). It is inferred from the figure that precedent approaches 1, 2, and 3 have better classified the 'US-News' and 'Middle-east' articles from the dataset. Still, the proposed fine-tuned CNN-LSTM model has shown better performance on different classes in the dataset. Finally, the performance of the proposed fine-tuned CNN-LSTM model is assessed using various standard metrics like Precision, Recall, F1 score, and Accuracy metrics. It can be observed from [Table 4](#) that for most of the classes in the dataset, the proposed fine-tuned CNN-LSTM model has achieved higher numerical values. The simulation results of the proposed model and epoch-wise training and validation metrics along with timing per step are presented in [Table](#)

5. Even though the present study has outperformed some precedent techniques and achieved higher accuracy and per-class PR, RC, and F1 scores, obtaining higher values for these metrics is considered a future research challenge using the same dataset. Moreover, the hardware constraint, i.e., working on a single CPU for implementing the proposed approach, resulted in low accuracy and other metrics values. This is another limited method with additional datasets and in GPU-enabled computers to see if it can improve the overall performance even further.

In the study, we have compared the performance of traditional ML models, advanced models, and DL-based transformer models like RNN, GRU over multiple datasets like LIAR, FakeNewsNet Politifact dataset (FNN-PF), (FakeNewsNet) Gossipcop (PNN-G), and Fake News Challenge (FNC) datasets concerning to the accuracy in [Table 6](#). LIAR is a sizable dataset of fake news that is freely accessible and was suggested in the study by W.Y. Wang [45]. In contrast to the FakeNewsNet dataset [46], a complete dataset comprising full-text news stories gathered from politifact.com and gossipcop.com, the LIAR dataset has roughly 12,800 entries. It consists mainly of user-profiles and brief political remarks. FakeNewsNet contains tagged news material such as news stories and social context data. The experimental findings of several research studies are deduced from [Table 6](#), and it shows that the proposed approach provides the best performance ratings across various fake news datasets.

Conventional machine learning classifiers are contrasted with modern ML methods like the Fine-tuned LSTM model and DL transformer-based pre-trained language models like BERT and RoBERTa. Devlin et al. first described Bidirectional Encoder Representation from Transformers [47], a DL model for various NLP tasks built on a transformer architecture. Preparing data and tokenization are the first steps in the BERT approach. In this stage, BERT tokenizes the text data into sub-word tokens and uses WordPiece tokenization. After tokenization, we require a BERT model that has already been trained, and we use this model to encode the tokenized text into numerical vectors.

For each token, contextual embedding is used, after which BERT is fine-tuned for tasks such as binary classification and model training. RoBERTa, known as Robustly Optimized BERT Approach is a modified version of the BERT strategy that Facebook first introduced. In this approach, by removing the next sentence prediction task from pre-training and training RoBERTa with 10 times as much data than BERT, this technique retrains BERT using a better training technique [48,49]. Advanced ML methods like LSTM perform substantially better when used to classify fake news on a binary scale, but in some instances, DL-based methods show superior improvements. The comparison of the metrics obtained by the researchers on different fake news datasets using traditional ML, advanced ML, and DL-based transformer models are mentioned below:

## 6. Conclusion

The news article classification is one of the promising fields from a research aspect. This research classifies news articles of longer sequential values using the proposed fine-tuned CNN-LSTM model. The model learns local and long-term dependencies from a sequence of data sets. The news articles were classified, and the categories of news articles were extracted from the dataset, enabling users to access the news of their interest. The experimental findings validate the superiority of

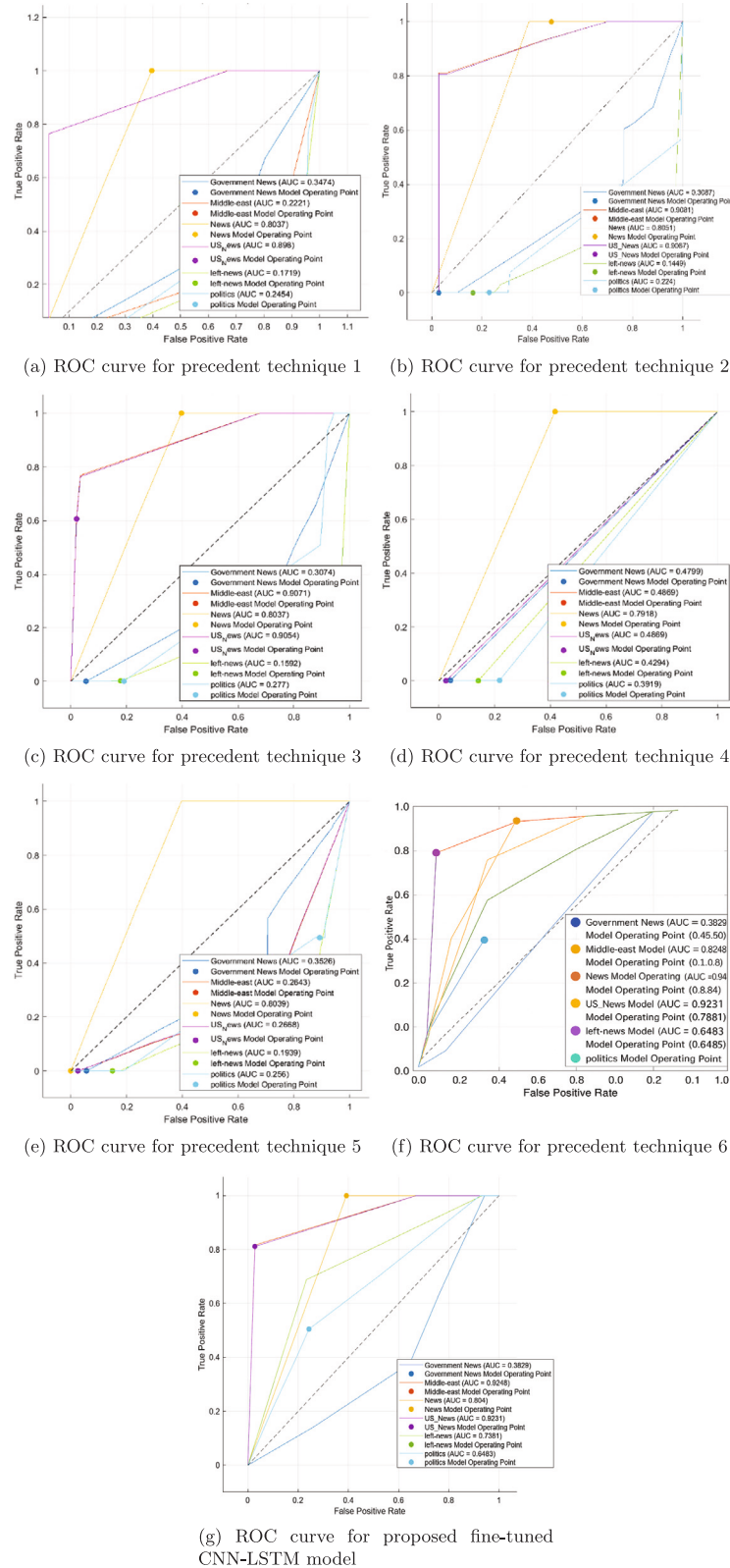


Fig. 6. Comparative analysis in terms of ROC curve between proposed fine-tuned CNN-LSTM model and precedent techniques.

the suggested method over the precedent techniques. Future work will involve classifying data using additional transformer-based techniques like BERT and XLNET to understand arbitrarily long documents and combining deep learning techniques with various datasets for better analysis and news classification with more iterations to increase the

preciseness of the classification outcome. These models enable complex transcriptions to be embedded, which facilitates the extraction of keywords. They have proven to be very capable of comprehending associations, context, and semantics inside text. With model improvements, accuracy in detecting false news can be increased. These models can

**Table 4**  
Comparative analysis in terms of various performance metrics between proposed fine-tuned CNN-LSTM model and precedent techniques.

| Models      | Precedent 1 |      |      |      | Precedent Approach 2 |      |      |      | Precedent Approach 3 |      |      |      | Precedent Approach 4 |      |      |      | Precedent Approach 5 |      |      |      | Precedent Approach 6 |      |      |      | Fine-Tuned CNN-LSTM |      |      |      |
|-------------|-------------|------|------|------|----------------------|------|------|------|----------------------|------|------|------|----------------------|------|------|------|----------------------|------|------|------|----------------------|------|------|------|---------------------|------|------|------|
|             | PR          | RC   | F1   | AC   | PR                   | RC   | F1   | AC   | PR                   | RC   | F1   | AC   | PR                   | RC   | F1   | AC   | PR                   | RC   | F1   | AC   | PR                   | RC   | F1   | AC   | PR                  | RC   | F1   | AC   |
| Govt News   | 0.10        | 0.12 | 0.11 |      | 0.00                 | 0.00 | 0.00 |      | 0.22                 | 0.05 | 0.08 |      | 0.00                 | 0.00 | 0.00 |      | 0.00                 | 0.00 | 0.00 |      | 0.21                 | 0.06 | 0.10 |      | 0.23                | 0.08 | 0.12 |      |
| Middle-east | 0.15        | 0.16 | 0.15 |      | 0.00                 | 0.00 | 0.00 |      | 0.11                 | 0.09 | 0.10 |      | 0.00                 | 0.00 | 0.00 |      | 0.00                 | 0.00 | 0.00 |      | 0.84                 | 0.91 | 0.87 |      | 0.42                | 0.25 | 0.31 |      |
| News        | 0.99        | 0.98 | 0.98 | 0.49 | 0.42                 | 1.00 | 0.59 | 0.41 | 0.99                 | 0.98 | 0.98 | 0.58 | 0.00                 | 0.00 | 0.00 | 0.31 | 0.40                 | 1.00 | 0.57 | 0.41 | 0.42                 | 0.45 | 0.44 | 0.66 | 0.88                | 0.94 | 0.91 | 0.68 |
| US-news     | 0.05        | 0.05 | 0.05 |      | 0.29                 | 0.07 | 0.11 |      | 0.17                 | 0.20 | 0.18 |      | 0.32                 | 0.06 | 0.10 |      | 0.30                 | 0.05 | 0.09 |      | 0.30                 | 0.16 | 0.21 |      | 0.45                | 0.48 | 0.47 |      |
| Left-News   | 0.14        | 0.18 | 0.15 |      | 0.00                 | 0.00 | 0.00 |      | 0.18                 | 0.12 | 0.14 |      | 0.00                 | 0.00 | 0.00 |      | 0.00                 | 0.00 | 0.00 |      | 0.39                 | 0.22 | 0.28 |      | 0.33                | 0.19 | 0.24 |      |
| Politics    | 0.33        | 0.26 | 0.29 |      | 0.41                 | 0.14 | 0.21 |      | 0.47                 | 0.67 | 0.55 |      | 0.31                 | 1.00 | 0.47 |      | 0.52                 | 0.13 | 0.21 |      | 0.45                 | 0.66 | 0.53 |      | 0.49                | 0.68 | 0.57 |      |

**Table 5**  
Epoch-wise training and validation metrics along with timing per step.

| Epoch | Time/Step | Loss   | Accuracy | Val_Loss |
|-------|-----------|--------|----------|----------|
| 1     | 395 ms    | 0.8811 | 0.6454   | 0.7777   |
| 2     | 395 ms    | 0.7671 | 0.6692   | 0.6712   |
| 3     | 1 s       | 0.6719 | 0.686    | 0.683    |
| 4     | 390 ms    | 0.6468 | 0.6916   | 0.6477   |
| 5     | 732 ms    | 0.6549 | 0.6869   | 0.7079   |
| 6     | 403 ms    | 0.6154 | 0.6967   | 0.6948   |
| 7     | 5 s       | 0.5834 | 0.7097   | 0.6776   |
| 8     | 466 ms    | 0.5644 | 0.7127   | 0.6998   |
| Test  | 765 ms    | 0.6998 | 0.6806   |          |

**Table 6**  
Comparative of the existing research studies.

| Reference | Techniques                                           | Dataset                                           | Measures      |
|-----------|------------------------------------------------------|---------------------------------------------------|---------------|
| [50]      | SVM, Random forest classifier                        | Modeled language markers and rhetorical relations | F-Score: 0.65 |
| [51]      | MLP using TF and TF-IDF                              | FNC dataset                                       | AC: 88.5      |
| [18]      | Linear SVM using TF and TF-IDF with unigram features | Sample of 2000 news pieces                        | AC: 92        |
| [52]      | RNN                                                  | LIAR                                              | AC: 21.5      |
|           | GRU                                                  |                                                   | AC: 21.7      |
|           | LSTM                                                 |                                                   | AC: 21.6      |
| [53]      | BERT                                                 | FNC dataset                                       | AC: 0.746     |
| [54]      | RoBERTa                                              | LIAR                                              | AC: 0.62      |
| [55]      | SVM                                                  | LIAR                                              | AC: 0.62      |
| [56]      | LR(n-gram)                                           | FNN-PF                                            | AC: 0.80      |
| [57]      | BiLSTM-BERT                                          | FNN-PF                                            | AC: 0.8558    |
| [58]      | CNN                                                  | FNN-G                                             | AC: 0.723     |
| [59]      | SVM                                                  | BuzzFeed News dataset                             | AC: 0.78      |

be adapted to new incorrect tactics and generalize well to other kinds of fake news as well. These approaches involve pre-training models on a large volume of data and then fine-tuning them for particular tasks. Integrating textual and visual data can yield a more thorough comprehension of the situation and raise the precision of classification overall, along with the explainability of the results.

CRediT authorship contribution statement

**Pratiyush Guleria:** Conceptualization, Data curation, Investigation, Methodology. **Jaroslav Frnda:** Formal analysis, Funding acquisition, Project administration, Resources, Validation. **Parvathaneni Naga Srinivasu:** Conceptualization, Software, Supervision, Validation, Writing – original draft, Writing – review & editing.

Ethical approval

The authors declare that they have no ethical approval required for this article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The dataset that is used in the current study is accessible as Fake and real news dataset in kaggle using the link <https://www.kaggle.com/datasets/clmentbisailon/fake-and-real-news-dataset>.

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